SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Fundamentals Of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO3: *Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 10 **Total Marks:100 marks**

Annexure A

POSTER PRESENTATION

Code of Conduct:

I M.Gayathri(192472074) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:M.Gayathri

RUBRICS FOR CALCULATION

Poster Presentation							
Criteria	**Weightage (%)**	**Level 4 (Exemplary) 5 Marks**	**Level 3 (Proficient) 4 Marks**	**Level 2 (Developing) 3 – 2 Marks**	**Level 1 (Beginning) 1 Marks**	**Level**	**Marks**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood		
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach		
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals		
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides		
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness		

Annexure A

Question 01: University Learning Management System (LMS) Usage Analysis (100 Marks)

A university uses an online Learning Management System (LMS) to deliver course materials. Most students spend **2–4 hours per week** on the platform, while a small number of students spend **more than 20 hours per week**. The university wants to understand student engagement patterns and identify factors affecting academic performance.

Prepare a comprehensive poster presentation that includes:

1. Data summarization of LMS usage statistics.
2. Analysis of the distribution of student engagement data.
3. Identification and interpretation of outliers.
4. Discussion on whether highly active students should be considered outliers.
5. Analysis of skewness and kurtosis in LMS usage patterns.
6. Relationship between LMS usage and academic performance.
7. Recommendations for improving student engagement and learning outcomes.

Question 02: Renewable Energy Production Analysis (100 Marks)

A solar power plant generates electricity throughout the year. On normal days, energy production ranges from **5,000 to 8,000 kWh**, while on exceptionally sunny days production exceeds **12,000 kWh**. During cloudy or rainy days, production may fall below **2,000 kWh**. The management wants to analyze production variability and improve energy forecasting.

Develop a poster presentation covering the following aspects:

1. Summarize the energy production data using descriptive statistics.
2. Examine the distribution of daily energy generation.
3. Identify and analyze unusually high and low production values.
4. Discuss appropriate methods for handling outliers.
5. Measure and interpret skewness and kurtosis.
6. Evaluate the impact of weather conditions on data distribution.
7. Suggest forecasting and operational strategies to improve renewable energy management.

Question 03: E-Commerce Festival Sales Analysis (100 Marks)

An online retailer records average daily sales of ₹10 lakh. During a festival season, sales increase to ₹2 crore per day.

Develop a poster discussing:

1. Summary statistics before and during the festival.
2. Changes in sales distribution.
3. Identification of unusual sales patterns.
4. Whether festival sales should be treated as outliers.
5. Interpretation of skewness and kurtosis.
6. Business insights obtained from the analysis.
7. Strategic recommendations for inventory planning.

Question 04: Healthcare Monitoring System (100 Marks)

A hospital continuously monitors patient heart rates using wearable sensors. Most readings are between 60–90 bpm, while some readings exceed 180 bpm.

Prepare a poster that addresses:

1. Data summarization of patient health records.
2. Visualization techniques for monitoring trends.
3. Detection of abnormal readings.
4. Methods for treating erroneous sensor values.
5. Analysis of skewness and kurtosis.
6. Differentiating medical emergencies from data errors.
7. Recommendations for real-time healthcare monitoring.

Question 05: Cybersecurity Network Traffic Analysis (100 Marks)

A company's server typically receives 20,000 requests per hour. During a cyberattack, the traffic suddenly increases to 2 million requests per hour.

Prepare a comprehensive poster explaining:

1. Methods for summarizing network traffic data.
2. Traffic distribution under normal conditions.
3. Detection of abnormal traffic spikes.
4. Outlier treatment strategies.
5. Analysis of skewness and kurtosis.
6. Role of anomaly detection in cybersecurity.
7. Security measures to prevent future attacks.

Question 06: Manufacturing Quality Control (100 Marks)

A manufacturing company produces 50,000 bottles daily. The target volume is 1 liter, but some bottles contain less than 900 ml or more than 1.1 liters.

Design a poster discussing:

1. Data summarization techniques.
2. Distribution of bottle volumes.
3. Detection of defective products using outlier analysis.
4. Treatment of abnormal measurements.
5. Analysis of skewness and kurtosis.
6. Impact on product quality and customer satisfaction.
7. Recommendations for process improvement.

Question 07: Climate Change and Weather Analysis (100 Marks)

A city normally experiences temperatures between 25°C and 35°C. Over the last decade, several days recorded temperatures above 45°C.

Prepare a poster presentation covering:

1. Summary statistics of temperature data.
2. Distribution of climatic variables.
3. Identification of extreme weather events.

4. Outlier treatment considerations.
5. Analysis of skewness and kurtosis.
6. Implications for climate change research.
7. Recommendations for policymakers.

Question 08: Social Media Engagement Analysis (100 Marks)

A content creator uploads 500 videos. Most videos receive fewer than 10,000 views, while a few viral videos receive over 10 million views.

Develop a poster explaining:

1. Data summarization methods.
2. Distribution of engagement metrics.
3. Detection of viral-content outliers.
4. Whether viral videos should be retained.
5. Interpretation of skewness and kurtosis.
6. Business implications for content creators.
7. Strategies to improve audience engagement.

Question 09: Stock Market Volatility Analysis (100 Marks)

A stock usually fluctuates within ±2% daily. Following a major corporate announcement, the stock price rises by 30% in a single day.

Prepare a detailed poster discussing:

1. Summary statistics of stock returns.
2. Distribution of financial data.
3. Detection of abnormal market movements.
4. Treatment of extreme observations.
5. Interpretation of skewness.
6. Importance of kurtosis in financial risk analysis.
7. Investment recommendations based on findings.

Question 10: Smart City Traffic Management System (100 Marks)

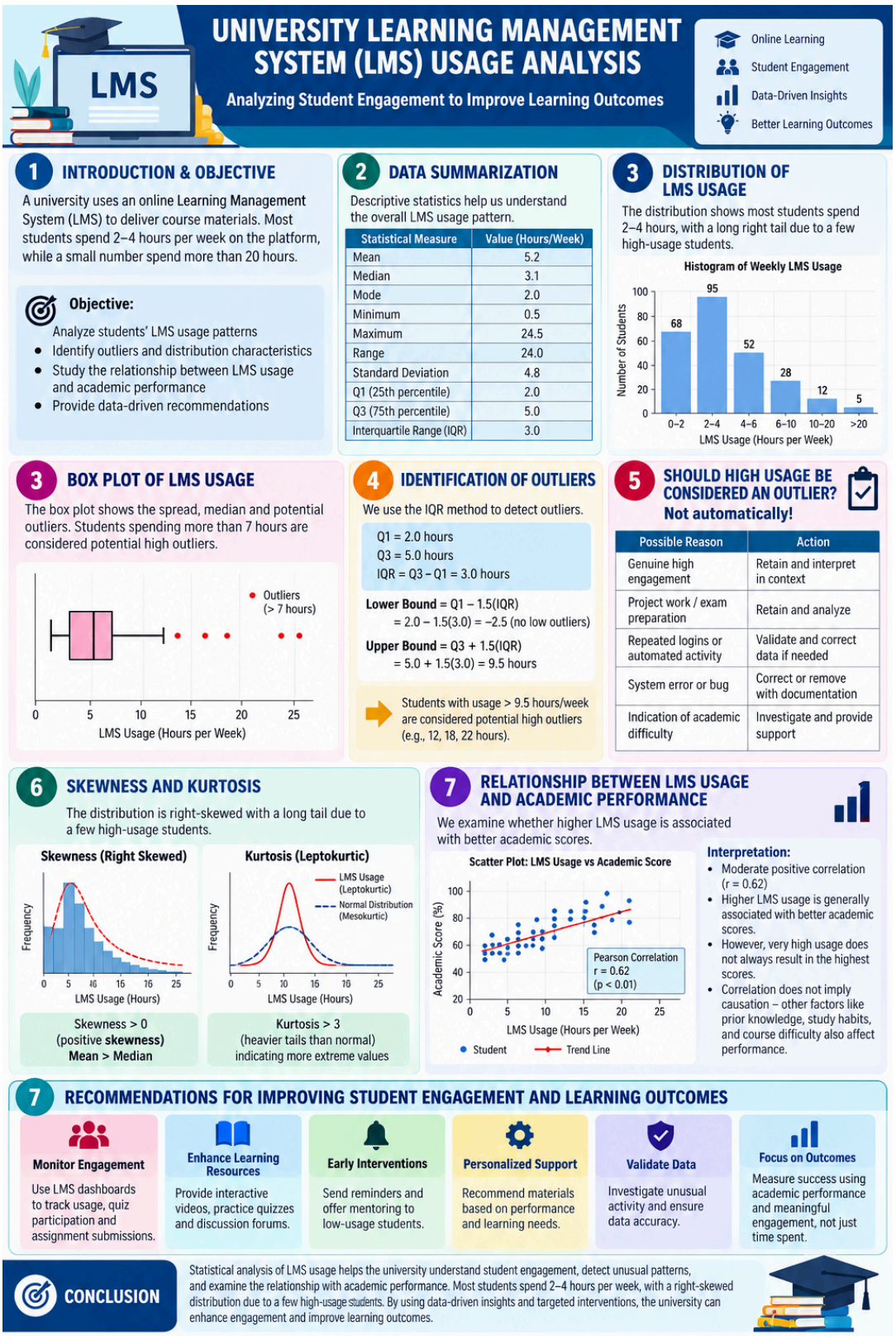
A smart city uses sensors to monitor traffic flow. The average traffic volume is 700 vehicles per hour. During a major cultural festival, traffic increases to 7,000 vehicles per hour.

Create a poster presentation that includes:

1. Data summarization of traffic patterns.
2. Analysis of traffic distribution.
3. Identification of unusual traffic events.
4. Outlier detection and treatment methods.
5. Analysis of skewness and kurtosis.
6. Traffic management recommendations.
7. Development of a predictive model for future events.

ANSWERS:

1.


LMS
UNIVERSITY LEARNING MANAGEMENT SYSTEM (LMS) USAGE ANALYSIS
Analyzing Student Engagement to Improve Learning Outcomes
Online Learning
Student Engagement
Data-Driven Insights
Better Learning Outcomes
1 INTRODUCTION & OBJECTIVE
A university uses an online Learning Management System (LMS) to deliver course materials. Most students spend 2–4 hours per week on the platform, while a small number spend more than 20 hours.
Objective:
Analyze students' LMS usage patterns
Identify outliers and distribution characteristics
Study the relationship between LMS usage and academic performance
Provide data-driven recommendations
2 DATA SUMMARIZATION
Descriptive statistics help us understand the overall LMS usage pattern.
Statistical Measure | Value (Hours/Week)
Mean | 5.2
Median | 3.1
Mode | 2.0
Minimum | 0.5
Maximum | 24.5
Range | 24.0
Standard Deviation | 4.8
Q1 (25th percentile) | 2.0
Q3 (75th percentile) | 5.0
Interquartile Range (IQR) | 3.0
3 DISTRIBUTION OF LMS USAGE
The distribution shows most students spend 2–4 hours, with a long right tail due to a few high-usage students.
Histogram of Weekly LMS Usage
Number of Students
68
95
52
28
12
5
0–2
2–4
4–6
6–10
10–20
>20
LMS Usage (Hours per Week)
3 BOX PLOT OF LMS USAGE
The box plot shows the spread, median and potential outliers. Students spending more than 7 hours are considered potential high outliers.
Outliers (> 7 hours)
LMS Usage (Hours per Week)
4 IDENTIFICATION OF OUTLIERS
We use the IQR method to detect outliers.
Q1 = 2.0 hours
Q3 = 5.0 hours
IQR = Q3 – Q1 = 3.0 hours
Lower Bound = Q1 – 1.5(IQR)
= 2.0 – 1.5(3.0) = –2.5 (no low outliers)
Upper Bound = Q3 + 1.5(IQR)
= 5.0 + 1.5(3.0) = 9.5 hours
Students with usage > 9.5 hours/week are considered potential high outliers (e.g., 12, 18, 22 hours).
5 SHOULD HIGH USAGE BE CONSIDERED AN OUTLIER? Not automatically!
Possible Reason | Action
Genuine high engagement | Retain and interpret in context
Project work / exam preparation | Retain and analyze
Repeated logins or automated activity | Validate and correct data if needed
System error or bug | Correct or remove with documentation
Indication of academic difficulty | Investigate and provide support
6 SKEWNESS AND KURTOSIS
The distribution is right-skewed with a long tail due to a few high-usage students.
Skewness (Right Skewed)
Frequency
LMS Usage (Hours)
Kurtosis (Leptokurtic)
LMS Usage (Leptokurtic)
Normal Distribution (Mesokurtic)
Frequency
LMS Usage (Hours)
Skewness > 0
(positive skewness)
Mean > Median
Kurtosis > 3
(heavier tails than normal)
indicating more extreme values
7 RELATIONSHIP BETWEEN LMS USAGE AND ACADEMIC PERFORMANCE
We examine whether higher LMS usage is associated with better academic scores.
Scatter Plot: LMS Usage vs Academic Score
Academic Score (%)
Pearson Correlation
r = 0.62
(p < 0.01)
LMS Usage (Hours per Week)
Student
Trend Line
Interpretation:
Moderate positive correlation (r = 0.62)
Higher LMS usage is generally associated with better academic scores.
However, very high usage does not always result in the highest scores.
Correlation does not imply causation – other factors like prior knowledge, study habits, and course difficulty also affect performance.
7 RECOMMENDATIONS FOR IMPROVING STUDENT ENGAGEMENT AND LEARNING OUTCOMES
Monitor Engagement
Use LMS dashboards to track usage, quiz participation and assignment submissions.
Enhance Learning Resources
Provide interactive videos, practice quizzes and discussion forums.
Early Interventions
Send reminders and offer mentoring to low-usage students.
Personalized Support
Recommend materials based on performance and learning needs.
Validate Data
Investigate unusual activity and ensure data accuracy.
Focus on Outcomes
Measure success using academic performance and meaningful engagement, not just time spent.
CONCLUSION
Statistical analysis of LMS usage helps the university understand student engagement, detect unusual patterns, and examine the relationship with academic performance. Most students spend 2–4 hours per week, with a right-skewed distribution due to a few high-usage students. By using data-driven insights and targeted interventions, the university can enhance engagement and improve learning outcomes.

2.

RENEWABLE ENERGY PRODUCTION ANALYSIS

A DATA-DRIVEN APPROACH TO UNDERSTAND ENERGY GENERATION AND IMPROVE SUSTAINABILITY

Clean Energy
Data Analysis
Efficient Planning
Sustainable Future

1 INTRODUCTION & OBJECTIVE

Renewable energy sources such as solar, wind and hydro play a key role in meeting the growing demand for clean and sustainable energy. The dataset contains daily/monthly energy production values from these sources.

Objective:

- Analyze renewable energy production data
- Identify distribution patterns and outliers
- Compare energy generation across sources
- Understand trends and seasonality
- Provide data-driven insights for better planning and sustainability

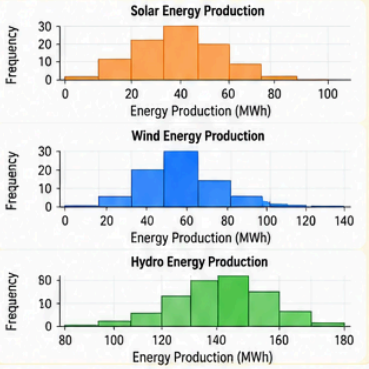
2 DATA SUMMARIZATION

Descriptive statistics help us understand the overall energy production pattern for each renewable source.

Statistical Measure (Sample Data)			
Statistic	**Solar (MWh)**	**Wind (MWh)**	**Hydro (MWh)**
Mean	52.4	68.1	120.5
Median	50.2	65.3	118.0
Mode	45.0	60.0	110.0
Minimum	12.0	20.0	85.0
Maximum	98.0	150.0	180.0
Range	86.0	130.0	95.0
Std. Deviation	18.6	27.3	22.5
Q1 (25%)	38.0	48.0	100.0
Q3 (75%)	65.0	85.0	140.0
IQR	27.0	37.0	40.0

3 DISTRIBUTION OF ENERGY PRODUCTION

Histograms show the distribution of energy generation for each renewable source.



4 IDENTIFICATION OF OUTLIERS

We use the IQR method to detect outliers.

IQR = Q3 − Q1
Lower Bound = Q1 − 1.5(IQR)
Upper Bound = Q3 + 1.5(IQR)

Energy Source	Q1	Q3	IQR	Lower Bound	Upper Bound
Solar	38.0	65.0	27.0	−2.5	105.5
Wind	48.0	85.0	37.0	−7.5	140.5
Hydro	100.0	140.0	40.0	40.0	200.0

Detected Outliers (Sample)

- Solar: values > 105.5 MWh (e.g., 110, 125 MWh)
- Wind: values > 140.5 MWh (e.g., 145, 150 MWh)
- Hydro: no high outliers in this sample (upper bound 200)

5 SHOULD HIGH PRODUCTION BE CONSIDERED AN OUTLIER?

Not automatically!

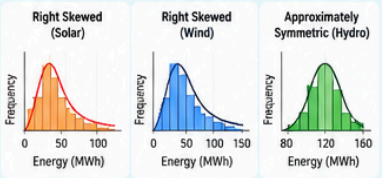
High energy production can be due to favorable weather conditions, seasonal variation or improved efficiency.

Possible Reason	Action
Genuine high production due to good weather (sunny/windy season)	Retain and interpret in context
Maintenance or sensor error	Validate and correct data if needed
One-time unusual event (storm, high wind)	Retain and analyze separately
Data entry mistake	Correct or remove with documentation
Shows an opportunity for better planning	Use to improve forecasting and grid management

6 SKEWNESS AND KURTOSIS

These measures help us understand the shape of the energy production distribution.

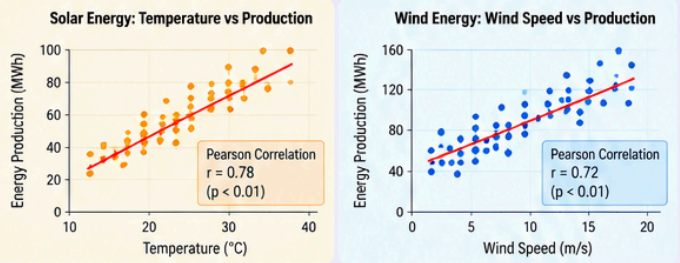
Energy Source	Skewness	Kurtosis (Excess)
Solar	0.85	1.20
Wind	0.62	0.75
Hydro	−0.25	−0.40



- Solar and wind show positive skewness (long right tail) due to a few high production days.
- Hydro is closer to symmetric distribution.

7 RELATIONSHIP BETWEEN ENERGY PRODUCTION AND FACTORS

We analyze how renewable energy production is related to external factors such as temperature (for solar) and wind speed (for wind) using correlation and regression.



- Strong positive correlation between temperature and solar energy production.
- Strong positive correlation between wind speed and wind energy production.
- Regression can be used to predict energy production based on weather variables.

RECOMMENDATIONS FOR IMPROVING RENEWABLE ENERGY PRODUCTION

1. Optimize resource utilization
Use forecasting models to plan solar and wind energy production based on weather predictions.

2. Improve infrastructure
Invest in better solar panels, wind turbines and energy storage systems to handle high and low production periods.

3. Monitor and maintain systems
Regularly check equipment and sensors to avoid data errors and ensure consistent production.

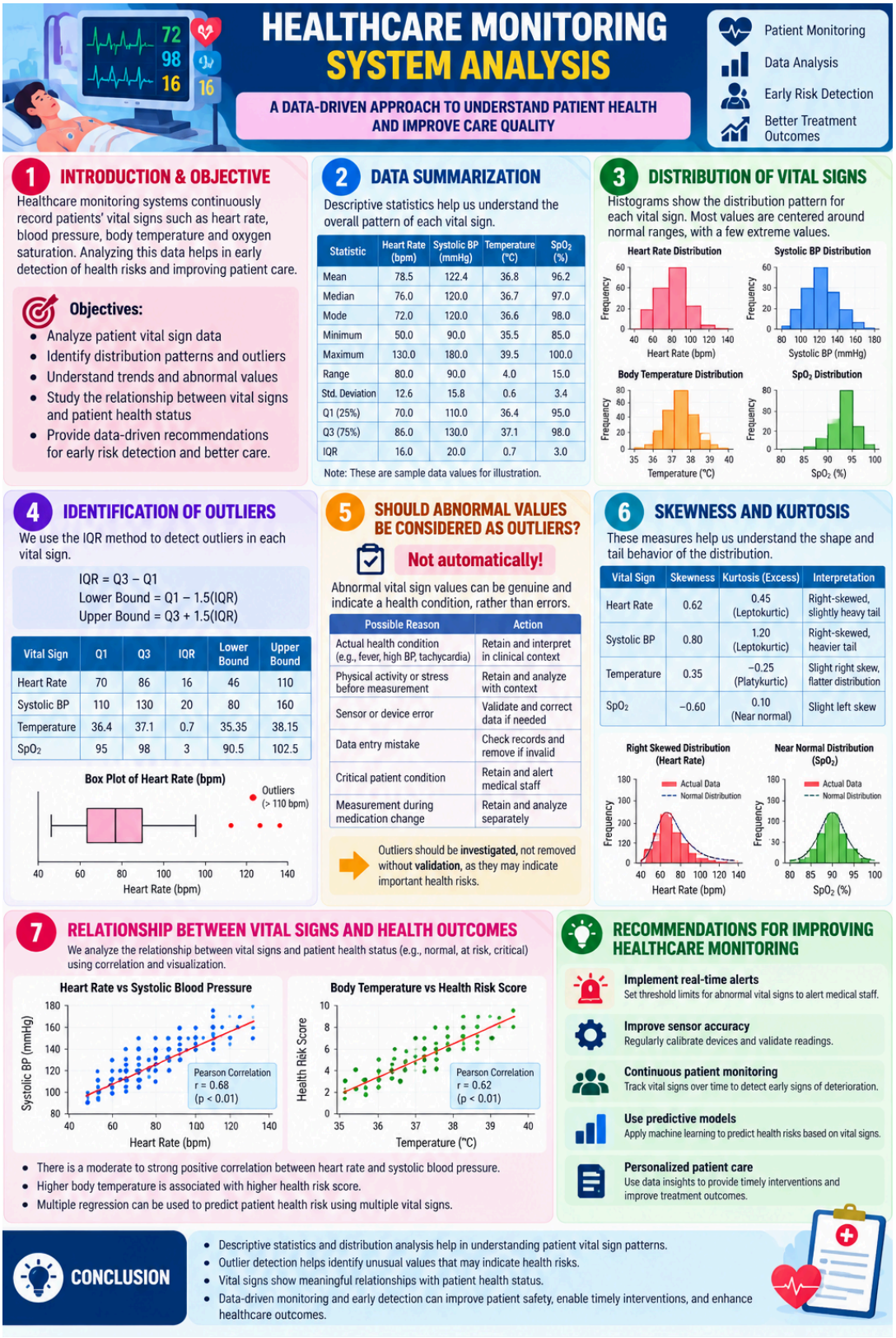
4. Use data-driven planning
Analyze historical data to identify high-production seasons and plan grid integration effectively.

5. Support sustainable policies
Encourage renewable energy adoption through incentives and long-term planning for a greener future.

CONCLUSION

- Descriptive statistics, distribution analysis and outlier detection help in understanding renewable energy production patterns.
- Solar and wind production are right-skewed with a few high-production days, while hydro is relatively stable.
- There is a strong positive relationship between weather factors (temperature, wind speed) and energy production.
- Data-driven insights can improve forecasting, resource planning and sustainability, contributing to a cleaner and greener future.

3.


HEALTHCARE MONITORING SYSTEM ANALYSIS
A DATA-DRIVEN APPROACH TO UNDERSTAND PATIENT HEALTH AND IMPROVE CARE QUALITY
Patient Monitoring
Data Analysis
Early Risk Detection
Better Treatment Outcomes
1 INTRODUCTION & OBJECTIVE
Healthcare monitoring systems continuously record patients' vital signs such as heart rate, blood pressure, body temperature and oxygen saturation. Analyzing this data helps in early detection of health risks and improving patient care.
Objectives:
Analyze patient vital sign data
Identify distribution patterns and outliers
Understand trends and abnormal values
Study the relationship between vital signs and patient health status
Provide data-driven recommendations for early risk detection and better care.
2 DATA SUMMARIZATION
Descriptive statistics help us understand the overall pattern of each vital sign.
Statistic | Heart Rate (bpm) | Systolic BP (mmHg) | Temperature (°C) | SpO2 (%)
Mean | 78.5 | 122.4 | 36.8 | 96.2
Median | 76.0 | 120.0 | 36.7 | 97.0
Mode | 72.0 | 120.0 | 36.6 | 98.0
Minimum | 50.0 | 90.0 | 35.5 | 85.0
Maximum | 130.0 | 180.0 | 39.5 | 100.0
Range | 80.0 | 90.0 | 4.0 | 15.0
Std. Deviation | 12.6 | 15.8 | 0.6 | 3.4
Q1 (25%) | 70.0 | 110.0 | 36.4 | 95.0
Q3 (75%) | 86.0 | 130.0 | 37.1 | 98.0
IQR | 16.0 | 20.0 | 0.7 | 3.0
Note: These are sample data values for illustration.
3 DISTRIBUTION OF VITAL SIGNS
Histograms show the distribution pattern for each vital sign. Most values are centered around normal ranges, with a few extreme values.
Heart Rate Distribution
Frequency
Heart Rate (bpm)
Systolic BP Distribution
Systolic BP (mmHg)
Body Temperature Distribution
Temperature (°C)
SpO2 Distribution
SpO2 (%)
4 IDENTIFICATION OF OUTLIERS
We use the IQR method to detect outliers in each vital sign.
IQR = Q3 − Q1
Lower Bound = Q1 − 1.5(IQR)
Upper Bound = Q3 + 1.5(IQR)
Vital Sign | Q1 | Q3 | IQR | Lower Bound | Upper Bound
Heart Rate | 70 | 86 | 16 | 46 | 110
Systolic BP | 110 | 130 | 20 | 80 | 160
Temperature | 36.4 | 37.1 | 0.7 | 35.35 | 38.15
SpO2 | 95 | 98 | 3 | 90.5 | 102.5
Box Plot of Heart Rate (bpm)
Outliers (> 110 bpm)
Heart Rate (bpm)
5 SHOULD ABNORMAL VALUES BE CONSIDERED AS OUTLIERS?
Not automatically!
Abnormal vital sign values can be genuine and indicate a health condition, rather than errors.
Possible Reason | Action
Actual health condition (e.g., fever, high BP, tachycardia) | Retain and interpret in clinical context
Physical activity or stress before measurement | Retain and analyze with context
Sensor or device error | Validate and correct data if needed
Data entry mistake | Check records and remove if invalid
Critical patient condition | Retain and alert medical staff
Measurement during medication change | Retain and analyze separately
Outliers should be investigated, not removed without validation, as they may indicate important health risks.
6 SKEWNESS AND KURTOSIS
These measures help us understand the shape and tail behavior of the distribution.
Vital Sign | Skewness | Kurtosis (Excess) | Interpretation
Heart Rate | 0.62 | 0.45 (Leptokurtic) | Right-skewed, slightly heavy tail
Systolic BP | 0.80 | 1.20 (Leptokurtic) | Right-skewed, heavier tail
Temperature | 0.35 | −0.25 (Platykurtic) | Slight right skew, flatter distribution
SpO2 | −0.60 | 0.10 (Near normal) | Slight left skew
Right Skewed Distribution (Heart Rate)
Actual Data
Normal Distribution
Frequency
Heart Rate (bpm)
Near Normal Distribution (SpO2)
SpO2 (%)
7 RELATIONSHIP BETWEEN VITAL SIGNS AND HEALTH OUTCOMES
We analyze the relationship between vital signs and patient health status (e.g., normal, at risk, critical) using correlation and visualization.
Heart Rate vs Systolic Blood Pressure
Systolic BP (mmHg)
Heart Rate (bpm)
Pearson Correlation
r = 0.68
(p < 0.01)
Body Temperature vs Health Risk Score
Health Risk Score
Temperature (°C)
Pearson Correlation
r = 0.62
(p < 0.01)
There is a moderate to strong positive correlation between heart rate and systolic blood pressure.
Higher body temperature is associated with higher health risk score.
Multiple regression can be used to predict patient health risk using multiple vital signs.
RECOMMENDATIONS FOR IMPROVING HEALTHCARE MONITORING
Implement real-time alerts
Set threshold limits for abnormal vital signs to alert medical staff.
Improve sensor accuracy
Regularly calibrate devices and validate readings.
Continuous patient monitoring
Track vital signs over time to detect early signs of deterioration.
Use predictive models
Apply machine learning to predict health risks based on vital signs.
Personalized patient care
Use data insights to provide timely interventions and improve treatment outcomes.
CONCLUSION
Descriptive statistics and distribution analysis help in understanding patient vital sign patterns.
Outlier detection helps identify unusual values that may indicate health risks.
Vital signs show meaningful relationships with patient health status.
Data-driven monitoring and early detection can improve patient safety, enable timely interventions, and enhance healthcare outcomes.

E-COMMERCE FESTIVAL SALES ANALYSIS

A DATA-DRIVEN APPROACH TO UNDERSTAND SALES PATTERNS AND IMPROVE BUSINESS STRATEGIES

- Data Insights
- Customer Trends
- Business Strategy
- Higher Revenue
- Better Customer Experience

1 INTRODUCTION & OBJECTIVE

E-commerce platforms experience a significant increase in sales during festival seasons such as Diwali, Big Billion Days and Great Indian Festival. This analysis focuses on understanding sales patterns, identifying outliers and studying factors that influence sales performance.

Objectives:

- Analyze e-commerce festival sales data
- Identify distribution patterns and outliers
- Study the relationship between key factors and sales
- Provide data-driven recommendations to improve future sales performance

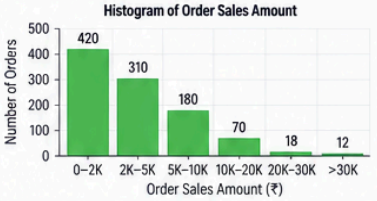
2 DATA SUMMARIZATION

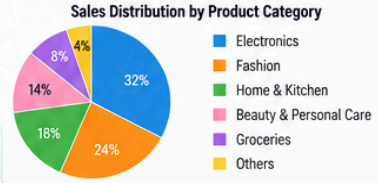
Descriptive statistics help us understand the overall sales pattern during the festival period.

Statistical Measure	Value (Sample Data)
Mean (Sales Amount ₹)	7,850
Median (₹)	5,200
Mode (₹)	3,000
Minimum (₹)	500
Maximum (₹)	48,000
Range (₹)	47,500
Standard Deviation (₹)	8,950
Q1 (25th percentile) (₹)	2,800
Q3 (75th percentile) (₹)	9,200
Interquartile Range (IQR)	6,400

3 DISTRIBUTION OF FESTIVAL SALES

The sales distribution is right-skewed, with most orders in the lower range and a few very high sales during peak days.

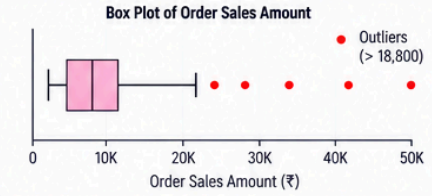




4 IDENTIFICATION OF OUTLIERS

We use the IQR method to detect outliers in sales amount.

IQR = Q3 – Q1 = 9,200 – 2,800 = 6,400
Lower Bound = Q1 – 1.5(IQR) = 2,800 – 9,600 = –6,800
Upper Bound = Q3 + 1.5(IQR) = 9,200 + 9,600 = 18,800



Detected Outliers (Sample)

- Orders with sales amount > ₹18,800 are considered high outliers (e.g., ₹22,000, ₹35,000, ₹48,000).
- These may be due to bulk purchases, expensive items (e.g., laptops, smartphones) or special festival offers.
- Outliers should be investigated, not removed without validation.

5 SHOULD HIGH SALES BE CONSIDERED AN OUTLIER?

Not automatically!

High sales values can be genuine and may indicate:

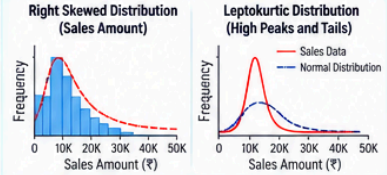
- Bulk purchases during festival offers
- High-value products (e.g., smartphones, laptops, appliances)
- Special discounts or limited-time deals
- Increased demand during peak days

Possible Reason	Action
Genuine customer purchase (high-value item)	Retain and interpret in context
Bulk order	Retain and analyze separately
Data entry error	Validate and correct data if needed
Fraudulent order or system issue	Investigate and remove if invalid
Special festival deal	Retain and analyze as a key business insight

6 SKEWNESS AND KURTOSIS

These measures help us understand the shape and tail behavior of the sales distribution.

Measure	Value (Sample Data)
Skewness	1.85 (Positive)
Kurtosis (Excess)	3.20 (Leptokurtic)



Interpretation:

- **Positive skewness:** Most orders are of lower value, with a few very high-value orders.
- **Leptokurtic:** Heavier tails than normal distribution, indicating more extreme high sales values.
- Mean is greater than median due to high-value orders.

7 RELATIONSHIP BETWEEN FACTORS AND SALES

We analyze the relationship between sales amount and factors such as discount percentage and customer ratings using correlation.

- There is a moderate to strong positive correlation between discount percentage and sales.
- Higher customer ratings are also associated with higher sales.
- Multiple regression can be used to predict sales based on factors like discount, rating, product category and day of sale.

RECOMMENDATIONS FOR IMPROVING FESTIVAL SALES

Optimize promotional strategies
Use targeted discounts and limited-time offers for high-demand categories.

Improve product visibility
Highlight top-rated products and best-selling items on the platform.

Enhance customer experience
Provide easy returns, fast delivery and personalized recommendations.

Inventory and supply planning
Use past sales data to predict demand and avoid stockouts.

Monitor and analyze sales trends
Use real-time dashboards to track sales, detect unusual patterns and plan future campaigns.

CONCLUSION

- Descriptive statistics, distribution analysis and outlier detection help us understand e-commerce festival sales patterns.
- Sales distribution is right-skewed with a few high-value orders.
- Discount percentage and customer ratings show a positive relationship with sales.
- Data-driven insights can help improve marketing strategies, inventory planning and overall business performance during future festivals.

5.

CYBERSECURITY NETWORK TRAFFIC ANALYSIS

A DATA-DRIVEN APPROACH TO DETECT ANOMALIES AND IMPROVE NETWORK SECURITY

- Network Security
- Anomaly Detection
- Data Analysis
- Threat Prevention
- Safer Digital Infrastructure

1 INTRODUCTION & OBJECTIVE

Network traffic data contains information about packets, protocols, source and destination IP addresses, and packet sizes. Analyzing this data helps in detecting unusual activities such as cyber attacks, traffic spikes, and malicious access attempts.

Objectives:

- Analyze network traffic data
- Identify distribution patterns and outliers
- Detect potential cyber attacks or anomalies
- Study the relationship between traffic features and attack occurrence
- Provide data-driven recommendations for stronger network security

2 DATA SUMMARIZATION

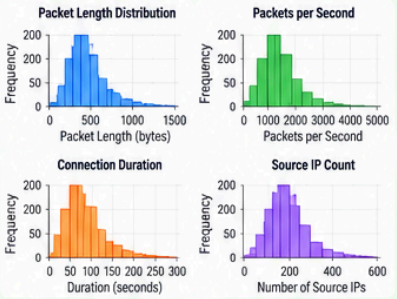
Descriptive statistics help us understand the overall pattern of network traffic features.

Statistical Measure (Sample Data)

Feature	Mean	Median	Std. Dev.	Min	Max
Packet Length (bytes)	450	420	220	40	1500
Packets per Second	1200	980	640	50	5000
Connection Duration (sec)	35.6	28.0	22.5	1	300
Source IP Count	320	310	85	50	650
Destination IP Count	280	260	70	45	600

3 DISTRIBUTION OF NETWORK TRAFFIC FEATURES

Histograms show the distribution pattern for key features. Most traffic is within normal range, with a few very high values.

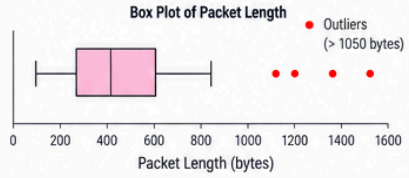


4 IDENTIFICATION OF OUTLIERS

We use the IQR method to detect outliers in each feature.

IQR = Q3 − Q1
Lower Bound = Q1 − 1.5(IQR)
Upper Bound = Q3 + 1.5(IQR)

Feature	Q1	Q3	IQR	Lower Bound	Upper Bound
Packet Length	300	600	300	−150	1050
Packets per Sec	700	1500	800	−500	2700
Connection Duration	10	60	50	−65	135
Source IP Count	200	400	200	−100	700
Destination IP Count	180	360	180	−90	630



5 SHOULD HIGH TRAFFIC VALUES BE CONSIDERED AN OUTLIER?

Not automatically!

High network traffic values may be genuine and can indicate:

- Legitimate high-usage periods (e.g., software updates, backups)
- Peak hours (e.g., online exams, events)
- Large file transfers
- Distributed traffic from many users

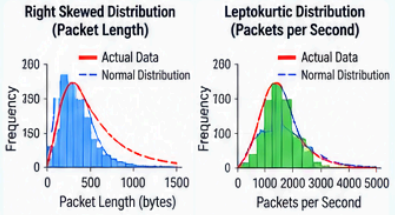
Possible Reason	Action
Genuine high network usage	Retain and interpret in context
Planned system maintenance	Retain and analyze separately
Cyber attack (e.g., DDoS, port scan)	Investigate and apply security measures
Anomalous traffic from unknown IP	Validate and check for malicious activity
Sensor or log error	Correct or remove with documentation

Outliers should be **investigated**, not removed without **validation**, as they may indicate important security events.

6 SKEWNESS AND KURTOSIS

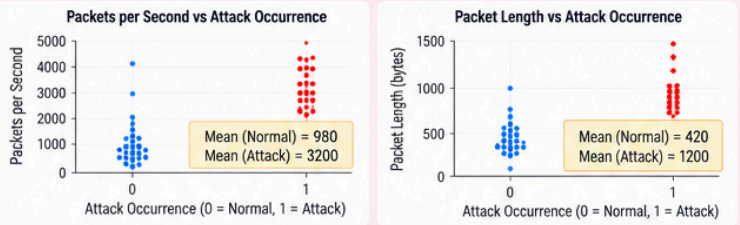
These measures help us understand the shape and tail behavior of the network traffic distribution.

Feature	Skewness	Kurtosis (Excess)	Interpretation
Packet Length	1.65	2.80	Right-skewed, heavy tail
Packets per Sec	1.40	1.95	Right-skewed, heavy tail
Connection Duration	1.10	0.85	Right-skewed
Source IP Count	0.65	0.20	Slight right skew
Destination IP Count	0.72	0.35	Slight right skew



7 RELATIONSHIP BETWEEN NETWORK TRAFFIC AND ATTACK OCCURRENCE

We analyze whether network traffic features can help predict attack occurrence using correlation and visualization.



- Higher packet length and packets per second are associated with a higher chance of attack.
- There is a strong positive correlation between network traffic features and attack occurrence.
- A classification model (e.g., logistic regression, decision tree) can be used to predict potential attacks.

RECOMMENDATIONS FOR IMPROVING NETWORK SECURITY

- **Implement real-time monitoring**
  Use anomaly detection to identify unusual traffic patterns.
- **Use traffic filtering and firewall rules**
  Block suspicious IP addresses and malicious traffic.
- **Plan for high-traffic events**
  Be prepared for legitimate traffic spikes during updates or special events.
- **Regularly analyze network logs**
  Detect early signs of attacks using data analytics.
- **Apply machine learning models**
  Use classification models to predict and prevent attacks.
- **Enhance security awareness**
  Educate users about safe practices to reduce attack risks.

CONCLUSION

- Descriptive statistics and distribution analysis help us understand normal and abnormal network traffic patterns.
- Outlier detection can reveal potential cyber attacks, but high values should be investigated before removal.
- Network traffic features show a strong relationship with attack occurrence.
- Data-driven monitoring and predictive models can improve network security and ensure a safer digital infrastructure.

6.

MANUFACTURING QUALITY CONTROL ANALYSIS

A DATA-DRIVEN APPROACH TO DETECT DEFECTS, ANALYZE PROCESS VARIATIONS AND IMPROVE PRODUCT QUALITY

- Quality Analysis
- Defect Detection
- Process Optimization
- Data-Driven Decisions

1 INTRODUCTION & OBJECTIVE

Manufacturing industries aim to produce high-quality products with minimal defects. Analyzing quality control data helps in identifying process variations, detecting outliers (defective items) and improving overall production efficiency.

Objectives:

- Analyze product quality data
- Identify distribution patterns and outliers
- Understand process variations
- Study the relationship between process parameters and defects
- Provide data-driven recommendations to improve product quality

2 DATA SUMMARIZATION

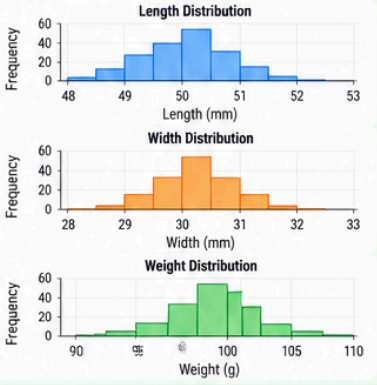
Descriptive statistics help us understand the overall quality characteristics of the products.

Statistical Measure	Length (mm)	Width (mm)	Weight (g)
Mean	50.02	30.01	100.5
Median	50.00	30.00	100.0
Mode	50.00	30.00	98.0
Minimum	48.5	28.2	92.0
Maximum	52.8	32.5	110.0
Range	4.3	4.3	18.0
Standard Deviation	0.85	0.92	2.80
Q1 (25th percentile)	49.4	29.3	98.0
Q3 (75th percentile)	50.6	30.7	102.5
IQR	1.2	1.4	4.5

Note: These are sample data values for illustration.

3 DISTRIBUTION OF PRODUCT MEASUREMENTS

Histograms show the distribution pattern for each measurement. Most products fall within the normal range, with a few extreme values.

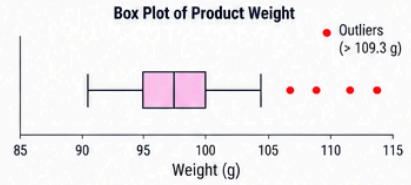


4 IDENTIFICATION OF OUTLIERS

We use the IQR method to detect outliers in each measurement.

IQR = Q3 − Q1
Lower Bound = Q1 − 1.5(IQR)
Upper Bound = Q3 + 1.5(IQR)

Feature	Q1	Q3	IQR	Lower Bound	Upper Bound
Length (mm)	49.4	50.6	1.2	47.6	52.4
Width (mm)	29.3	30.7	1.4	27.2	32.8
Weight (g)	98.0	102.5	4.5	91.3	109.3



5 SHOULD DEFECTIVE PRODUCTS BE CONSIDERED OUTLIERS?

Not automatically!

Extreme values may indicate defective products, but they could also be due to:

- Machine calibration issues
- Raw material variation
- Measurement errors
- Special production batches

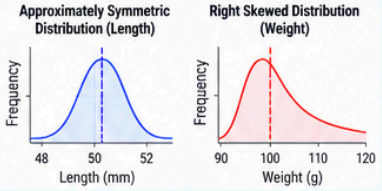
Possible Reason	Action
Defective product (out of specification)	Retain and classify as defect
Machine or process variation	Investigate and analyze separately
Measurement error	Validate data and correct if needed
Special batch or prototype	Retain and interpret in context

Outliers should be **investigated**, not removed without **validation**, as they may reveal important process issues.

6 SKEWNESS AND KURTOSIS

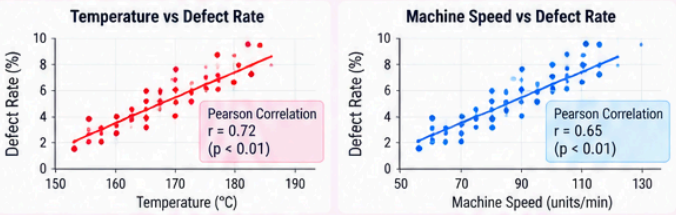
These measures help us understand the shape and tail behavior of the product measurements.

Feature	Skewness	Kurtosis (Excess)	Interpretation
Length	0.25	0.10	Approximately symmetric
Width	−0.30	0.45	Slight left skew
Weight	1.20	2.10	Right-skewed, heavy tail (more extreme values)



7 RELATIONSHIP BETWEEN PROCESS PARAMETERS AND DEFECT RATE

We analyze the relationship between manufacturing parameters and defect rate using correlation and regression analysis.



- There is a strong positive correlation between temperature and defect rate.
- Higher machine speed is also associated with an increase in defect rate.
- Multiple regression can be used to predict defect rate based on process parameters.

RECOMMENDATIONS FOR IMPROVING PRODUCT QUALITY

Optimize process parameters
Maintain optimal temperature and machine speed based on analysis to reduce defect rate.

Regular quality monitoring
Use control charts and real-time data analytics to detect anomalies early.

Machine maintenance
Perform regular calibration and maintenance to avoid process variations.

Use predictive models
Apply machine learning models to predict defects and enable preventive actions.

Continuous improvement
Analyze defect patterns and implement Six Sigma or quality management practices.

CONCLUSION

- Descriptive statistics and distribution analysis help us understand product quality and process variations.
- Outlier detection identifies potential defective products, but they should be investigated before removal.
- Process parameters such as temperature and machine speed show a strong relationship with defect rate.
- Data-driven insights can help in improving product quality, reducing defects, and enhancing manufacturing efficiency.

7.

QUESTION 7

CLIMATE CHANGE AND WEATHER ANALYSIS

Analyzing Temperature Trends and Extreme Heat Events

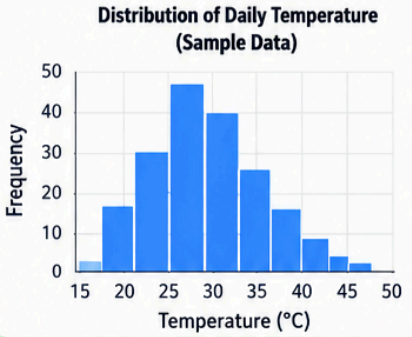
1 Objective

- Analyze temperature data to understand climate trends.
- Identify extreme heat events (> 45°C).
- Use descriptive statistics and distribution analysis.
- Provide insights and recommendations for climate adaptation.

2. Descriptive Statistics (Temperature in °C)

Statistic	Value (Sample Data)
Mean	32.5
Median	31.8
Mode	30.2
Minimum	18.4
Maximum	47.6
Standard Deviation	6.8
Q1	28.1
Q3	36.9
IQR	8.8

3. Distribution of Temperature



4. Outlier Detection (IQR Method)

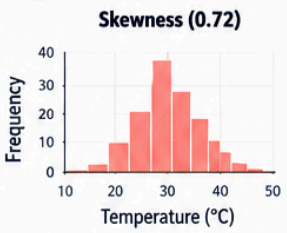
Q1 = 28.1°C
Q3 = 36.9°C
IQR = 8.8°C

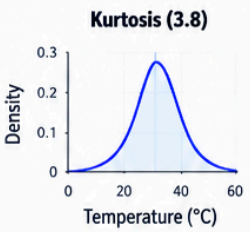
Lower Bound = Q1 – 1.5(IQR)
= 28.1 – 1.5(8.8)
= 14.9°C

Upper Bound = Q3 + 1.5(IQR)
= 36.9 + 1.5(8.8)
= 50.1°C

Temperatures > 50.1°C or < 14.9°C are potential outliers.

5. Skewness and Kurtosis





- Positive skewness (right-skewed).
- More high-temperature days.
- Mean > Median due to extreme heat neat events.

- Leptokurtic (heavy tails).
- More extreme temperature values than normal distribution.

6. Analysis of Extreme Heat Events

Year	Days > 45°C	Maximum (°C)	Interpretation
2020	3	45.8	Isolated events
2021	5	46.2	Increasing trend
2022	7	47.1	More frequent
2023	9	47.6	Significant rise

Insights

- Increase in frequency and intensity of extreme heat events.
- Possible impact of climate change.
- Requires long-term monitoring and mitigation.

7. Recommendations

- Improve green cover and urban planning (plants, parks, tree plantation).
- Implement heatwave early warning systems.
- Ensure water availability and cooling centers during extreme heat.
- Adopt climate-resilient infrastructure (e.g., cool roofs, heat-resistant materials).
- Increase public awareness about heat risks and preparedness.

Conclusion

- The temperature data shows a rising trend and more frequent extreme heat events, indicating the impact of climate change.
- Continuous monitoring and proactive strategies are essential for a sustainable and resilient future.
- Measures such as green infrastructure, early warnings, water management, and public awareness can help reduce the risks of extreme heat and improve climate adaptation.

8.

QUESTION 8

SOCIAL MEDIA ENGAGEMENT ANALYSIS

Understanding User Interaction and Content Performance

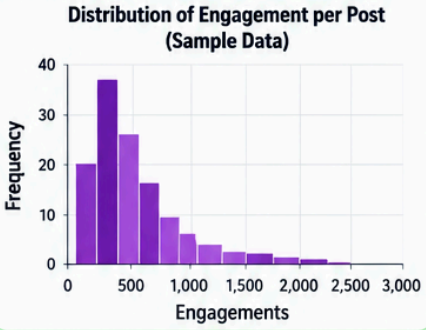
1 Objective

- Analyze user engagement data on social media platforms.
- Identify patterns and outliers in engagement metrics.
- Use descriptive statistics and distribution analysis.
- Provide insights and recommendations to improve content strategy.

2 Descriptive Statistics (Engagements per Post)

Statistic	Value (Sample Data)
Mean	520
Median	380
Mode	310
Minimum	50
Maximum	4,200
Standard Deviation	640
Q1	220
Q3	680
IQR	460

3 Distribution of Engagement



4 Outlier Detection (IQR Method)

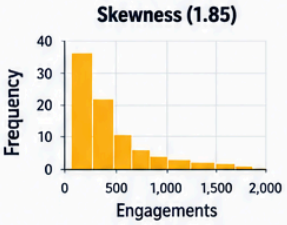
Q1 = 220
Q3 = 680
IQR = 460

Lower Bound = Q1 – 1.5(IQR)
= 220 – 1.5(460)
= –470 (set to 0)

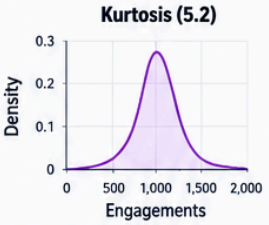
Upper Bound = Q3 + 1.5(IQR)
= 680 + 1.5(460)
= 1,370

Engagements > 1,370 are potential outliers (viral posts).

5 Skewness and Kurtosis



- Highly right-skewed distribution.
- Few posts get very high engagement.
- Most posts have low to medium engagement.



- Leptokurtic distribution (heavy tails).
- More extreme high engagement values than normal distribution.
- Indicates presence of viral posts.

6 Analysis of High Engagement Posts (Outliers)

Post ID	Engagements	Type	Interpretation
P15	2,450	Video	Informative & trending
P27	3,120	Meme	Viral content
P41	1,980	Live stream	High audience interest
P58	4,200	Announcement	Exceptional reach

- High engagement posts are not errors.
- They indicate successful content and audience interest.
- Can be used to identify effective content strategies.
- Helps in understanding what type of content works best (videos, memes, live streams, announcements).

7 Recommendations

- Focus on high-performing content types (videos, live sessions, interactive posts).
- Post at optimal times (when audience is most active).
- Analyze audience preferences (e.g., videos get higher engagement).
- Use data-driven content strategy.
- Monitor for fake engagement or bot activity.
- Collaborate with influencers and encourage user-generated content.

Conclusion

- Social media engagement data is right-skewed with high outliers (viral posts).
- These outliers provide valuable insights into successful content.
- A data-driven content strategy, optimal posting times, and audience analysis can improve reach and engagement.

9.

QUESTION 9

STOCK MARKET VOLATILITY ANALYSIS

Analyzing Daily Returns and Identifying Unusual Market Movements

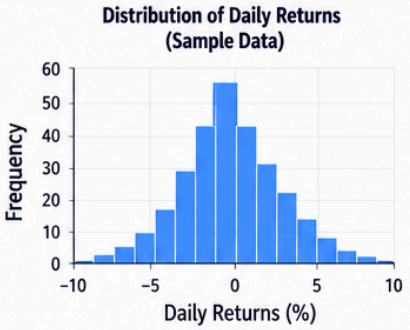
1 Objective

- Analyze daily stock returns to understand market behavior.
- Identify unusual market movements (high volatility).
- Use descriptive statistics and distribution analysis.
- Provide insights for risk management and investment decisions.

2 Descriptive Statistics (Daily Returns %)

Statistic	Value (Sample Data)
Mean	0.08
Median	0.05
Mode	0.02
Minimum	−8.4
Maximum	7.6
Standard Deviation	2.1
Q1	−0.9
Q3	1.2
IQR	2.1

3 Distribution of Daily Returns



4 Outlier Detection (IQR Method)

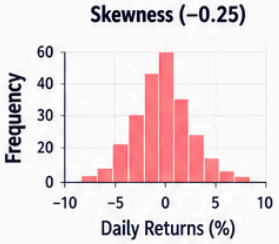
Q1 = −0.9%
Q3 = 1.2%
IQR = 2.1%

Lower Bound = Q1 − 1.5(IQR)
= −0.9 − 1.5(2.1)
= −4.05%

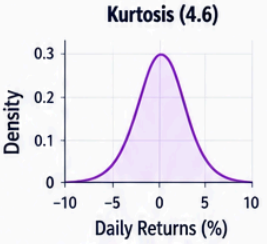
Upper Bound = Q3 + 1.5(IQR)
= 1.2 + 1.5(2.1)
= 4.35%

Daily returns < −4.05% or > 4.35% are potential outliers (high volatility days).

5 Skewness and Kurtosis



- Slightly left-skewed distribution.
- Indicates more extreme negative returns.
- Mean < Median.



- Leptokurtic distribution (heavy tails).
- More extreme values than normal distribution.
- Higher probability of large market movements.

6 Analysis of High Volatility Days

Date	Daily Return (%)	Event	Interpretation
15 Jan	−5.8	Global market sell-off	High volatility
12 Mar	6.4	Earnings surprise	Positive spike
23 Apr	−7.1	Geopolitical tensions	Sharp decline
18 Aug	5.9	Policy announcement	High volatility

- Extreme values are not necessarily errors.
- Often caused by significant market events (earnings, geopolitical issues, policy changes).
- Important for risk assessment and investment strategies.

7 Recommendations

- Monitor market trends and key economic events regularly.
- Use risk management strategies (e.g., stop-loss).
- Diversify investment portfolio to reduce risk.
- Apply statistical models (e.g., volatility clustering, ARIMA, GARCH) for prediction.
- Use historical data for better decision making and long-term investment planning.

Conclusion

- Stock market returns show heavy tails and occasional extreme movements.
- These outliers are important indicators of market volatility and are often linked to major market events.
- Statistical analysis helps investors make informed decisions and manage risk effectively.

10.

QUESTION 10

SMART CITY TRAFFIC MANAGEMENT

Analyzing Traffic Flow to Improve Urban Mobility

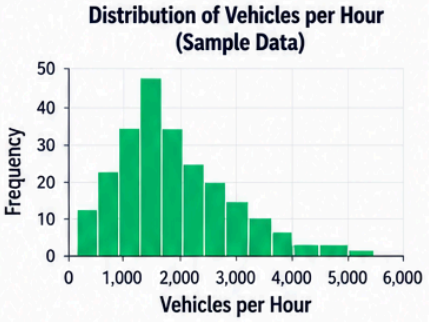
1 Objective

- Analyze vehicle count data from urban sensors.
- Identify peak traffic hours and unusual congestion.
- Use descriptive statistics and distribution analysis.
- Provide recommendations for better traffic management and urban mobility.

2 Descriptive Statistics (Vehicles per Hour)

Statistic	Value (Sample Data)
Mean	1,850
Median	1,620
Mode	1,400
Minimum	320
Maximum	5,800
Standard Deviation	920
Q1	1,100
Q3	2,300
IQR	1,200

3 Distribution of Traffic Flow



4 Outlier Detection (IQR Method)

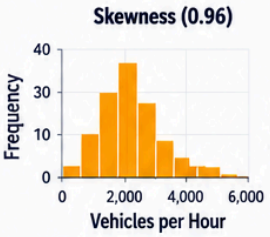
Q1 = 1,100
Q3 = 2,300
IQR = 1,200

Vehicle count > 4,100 is a potential outlier (severe congestion).

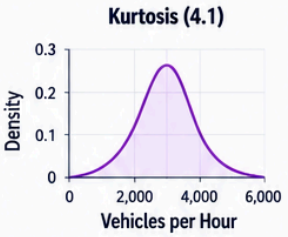
Lower Bound = Q1 – 1.5(IQR)
= 1,100 – 1.5(1,200)
= –700 (set to 0)

Upper Bound = Q3 + 1.5(IQR)
= 2,300 + 1.5(1,200)
= 4,100

5 Skewness and Kurtosis



- Right-skewed distribution.
- More high-traffic hours.
- Indicates accasional severe congestion.



- Leptokurtic distribution (heavy tails).
- More extreme congestion values than normal distribution.
- Higher probability of traffic spikes.

6 Analysis of Peak Traffic Hours

Time	Vehicle Count	Interpretation
7 AM	3,850	Morning peak (office hours)
9 AM	4,320	High congestion
1 PM	2,100	Moderate traffic
5 PM	5,200	Evening peak (highest)
10 PM	680	Low traffic

- Peak hours show significantly higher vehicle counts.
- High values are not errors; they indicate real congestion.
- Helps in planning and traffic signal optimization.

7 Recommendations

- Optimize traffic signal timing using real-time data.
- Promote public transportation.
- Encourage non-motorized travel (e.g., cycling, walking).
- Use real-time monitoring systems and IoT sensors.
- Implement smart traffic management using data analytics and AI.

Conclusion

- Traffic flow data is right-skewed with high outliers during peak hours.
- Statistical analysis helps identify congestion patterns and unusual traffic surges.
- Implementing smart traffic management strategies can improve urban mobility, reduce congestion and make cities more sustainable.